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The Use of Electrostatic Probes for Plasma Diagnostics—A Review

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The use of electrostatic, or Langmuir, probes for plasma diagnostics is reviewed. The emphasis is on experimental implementation and current techniques, and particular attention is paid to sources of error in theoretical interpretation as well as to experimental problems that can occur in complex, reactive plasmas.

KEY WORDS: Plasma diagnostics; electrostatic probes; Langmuir probes.

1. INTRODUCTION

The electrostatic or "Langmuir" probe is the most widely used form of plasma diagnostic presently in use. Indeed, a cursory review of journal article titles contained in the computerized INSPEC files has shown that there have been over 2000 articles dealing with probes in plasmas published within the last 12-year period alone. The popularity of probes for plasma diagnostics is not surprising. A superficial examination of probe theory and practice gives the seductive prospect of being able to deduce plasma potential, electron energy distribution function, and plasma density by simply introducing a small metallic wire into the plasma and by measuring the probe voltage–current characteristic with respect to a reference electrode in the plasma. A slightly closer look at electrostatic plasma probes reveals quite a different picture. The experimental procedures are still relatively simple, but do demand some care and attention to detail if meaningful data are to be obtained. The theory of probes, on the other hand, has a tendency to become extremely complicated if experimental probe characteristics are to be related with any precision and generality to the fundamental plasma characteristics that we wish to deduce. Moreover, as plasmas become more complex, as is definitely the case in virtually all plasma chemistry applications, a myriad of complicating factors affect both

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the theoretical interpretation and the experimental implementation of the measurements. The combination of easy experiment and easy theory, as is no surprise, remains an illusion.

The purpose of this paper is to review briefly those issues of electrostatic probe plasma diagnostics that are of primary interest to those working in the field of plasma chemistry. The emphasis will be primarily on experimental implementation and current techniques, rather than on theoretical interpretations; and particular attention will be paid to problems that occur in complex, reactive plasmas.

Fortunately, a number of excellent comprehensive reviews and evaluations of plasma diagnostics with probes are readily available, so a detailed exposition is not necessary. It is advisable, however, for anyone contemplating the use of electrostatic probes for plasma diagnostics to carefully review a number of these references so that a full appreciation of the subject can be obtained. In particular, the early exposition by Loeb,[1] with his recent update,[2] contains a particularly useful discussion of sources of error. The article by Chen[3] provides a nice blend of theory and experiment with a useful discussion of the effects of magnetic fields on probe characteristics. The monograph by Swift and Schwar[4] is thorough, particularly on the theory of probes. The monograph by Chung, Talbot, and Touryan[5] provides a valuable discussion of the use of probes in flowing plasmas, and the paper by Smy[6] provides an excellent review of the use of probes in high-pressure (atmospheric) plasmas.

2. BASIC CONCEPTS OF ELECTROSTATIC PROBE BEHAVIOR IN PLASMAS

Prior to discussing the behavior of an electrostatic probe in a plasma, it is necessary to define a "regime" of operation. This is done most conveniently by comparing the size of the probe, r_p, to the mean free path of the charged particles in the medium, l, and to the Debye length, λ_D. Figure 1 schematically defines the various probe regimes in terms of the ratios of these characteristic lengths.

The Debye length, λ_D, is the distance in a plasma over which significant deviations from charge neutrality, and therefore significant electrostatic fields, can exist. It defines the size of the primary sheath region surrounding the probe. This quantity varies inversely with the square root of the plasma density and directly as the square root of the electron energy. If the Debye length is much smaller than the mean free path ($l \gg \lambda_D$), particles crossing the sheath will suffer no collisions (collisionless sheath). If $l \ll \lambda_D$, the particles will suffer many collisions as they cross the sheath (collisional sheath).

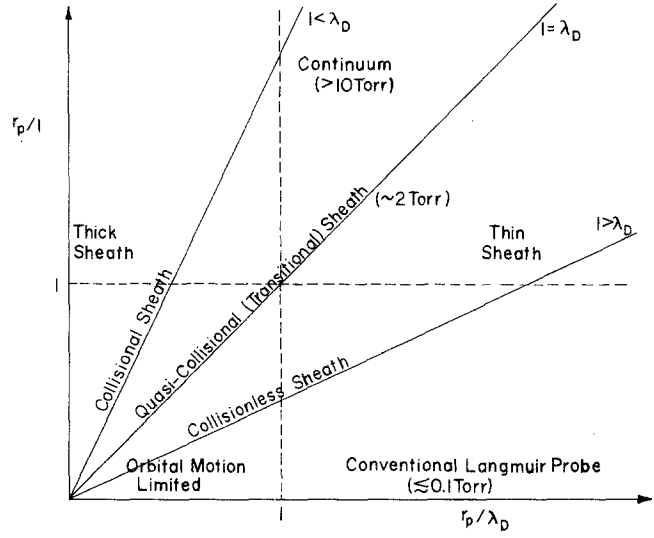


Fig. 1. Probe operation regimes in terms of probe size, r_p, mean free path, l, and Debye length, λ_D.

If the size of the probe is much smaller than the Debye length ($r_p/\lambda_D \ll 1$), the sheath is "thick" compared to the probe. If $r_p/\lambda_D \gg 1$ the sheath is "thin." (This is mainly relevant to spherical and cylindrical probes.)

If the pressure is very low so that the mean free path is large and $r_p/l \ll 1$, we are dealing with essentially collisionless systems characteristic of conventional Langmuir probe theory. In the high-pressure situation $r_p/l \gg 1$ and the continuum theories must be used.

In each situation, the appropriate ratios need to be evaluated in order to define the probe regime. However, as a crude guide, some approximate gas pressure ranges have been noted in Fig. 1 for some typical plasma chemistry regimes.

2.1. Probes in Low-Pressure Plasmas

The only probe regime that can be reasonably easily understood is that which corresponds to the conventional Langmuir probe theory, or the thin-sheath collisionless regime $l \gg r_p \gg \lambda_D$ for low-pressure plasmas.

Consider the V–I characteristic of a small metallic probe immersed in a low-pressure plasma with the return current passing through a large reference electrode as shown in Fig. 2. The current vs. applied voltage is typically as shown in Fig. 3. There are basically three distinct regions of

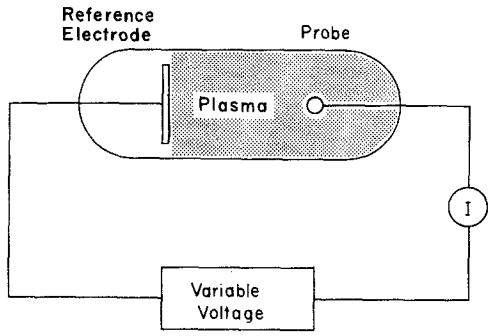


Fig. 2. Typical probe measuring circuit.

the characteristic. Let us begin the discussion at the point where the applied voltage equals the plasma potential, V_p. This a fundamental quantity of interest in defining the nature of the plasma with respect to the bounding surfaces. By definition, when the probe is at V_p, it is at the same potential as the surrounding plasma so there is neither a retarding nor an accelerating field acting on the ions or electrons in the vicinity of the probe. The current to the probe is then determined by the capture cross section, or geometric cross section in this case, of the probe and the random flux of particles incident on the probe. If for the moment we consider a plasma consisting of equal numbers of electrons and positive ions and that both have a Maxwellian energy distribution defined by a temperature T, then the electron current is

$$I_e = \tfrac{1}{4} A_p ne (8kT_e/\pi m)^{1/2} \tag{1}$$

and the ion current is

$$I_+ = \tfrac{1}{4} A_p ne (8kT_+/\pi M)^{1/2} \tag{2}$$

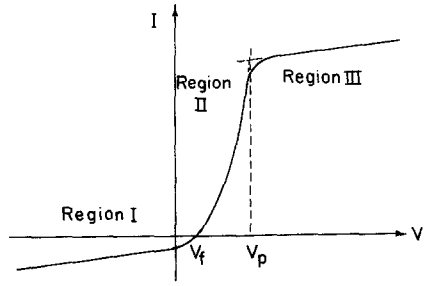


Fig. 3. Typical probe I–V characteristic.

where n is the plasma density, T_e and T_+ are the electron and ion temperatures respectively, m and M are the electron and ion masses, e is the electron and ion charge, and A_p is the collection area of the probe. Of course $M \gg m$ and, in low-pressure plasmas, usually $T_e \gg T_+$; so there is a considerable imbalance in electron and ion flux to the probe. Unfortunately, this complicates the experimental problem of precisely defining the point at which $V = V_p$. As can be seen, the potential achieved by the probe when no current is shown, the floating potential, V_f, is significantly different from the plasma potential, so the convenient reference point of zero current cannot be used to determine V_p. What happens here, as designated by region II in Fig. 3, is that the probe potential is driven negative with respect to the plasma potential so that a retarding field is set up for the electrons and an accelerating field is set up for positive ions. Again, if the electron distribution function is Maxwellian, the electron current is given by

$$I_e = \tfrac{1}{4} A_p n e (8kT_e/\pi m)^{1/2} \exp(-eV_\phi/kT_e) \tag{3}$$

where $V_\phi = V_p - V$ is the retarding potential. Only when the retarding potential is sufficiently large, i.e., $V_\phi \equiv V_p - V_f$, so that the electron and ion flux are equal, does the current null out and the floating potential is reached. Under the assumption that the ion current is small for $V > V_f$, the simplest interpretation of the probe characteristic in region II is given simply by Eq. 3. Obviously, a plot of $\ln I_e$ vs. V_ϕ will then give a direct measure of the electron temperature T_e, one of the fundamental parameters.

In the case where the distribution function is not Maxwellian but is still isotropic, which is really more typical, the electron current is

$$I_e(V_\phi) = neA_p(2\pi e/m^2) \int_{eV_\phi}^{\infty} \varepsilon f(\varepsilon)(1 - eV_\phi/\varepsilon)\, d\varepsilon \tag{4}$$

where $\varepsilon = \frac{1}{2}mv^2$ is the electron energy and $f(\varepsilon)$ is the isotropic electron energy distribution function normalized so that a Maxwellian distribution would be

$$f(\varepsilon) = (m/2\pi kT_e)^{3/2} \exp(-\varepsilon/kT_e) \tag{5}$$

In this case

$$d^2I_e/dV_\phi^2 = (neA_p 2\pi e^3/m^2) f(eV_\phi) \tag{6}$$

so that the distribution function, $f(\varepsilon)$, can be deduced directly from the second derivative of the probe characteristic

$$f(\varepsilon) = \frac{m^2}{neA_p 2\pi e^3} \left. \frac{d^2 I_e(V_\phi)}{dV_\phi^2} \right|_{\varepsilon = eV_\phi} \tag{7}$$

This technique for determining the distribution function is widely used and will be discussed later.

Now let us turn to a discussion of region III, where $V > V_p$. In this region a space-charge field is set up that accelerates electrons toward the probe and repels ions. Since in this case we have been assuming that the ion current contribution is negligible, decreasing the ion flux should have relatively little effect on the current. Increasing V should also have relatively little effect on the electron current since one would think that it would be limited by the random electron flux entering the accelerating field of the probe. In this sense, we would expect region III to represent a region of electron current saturation where the current is given approximately by Eq. (1). Unfortunately, life is not quite that simple. The fact that $V \neq V_p$ implies the existence of a region of charge imbalance, or "sheath," in which an electric field exists surrounding the probe. Changing the potential applied to the probe will have some effect on the penetration distance into the plasma. If the probe is spherical or cylindrical, varying V will vary the *capture* area of the probe since this is given by the outer dimensions of the *sheath* upon which the random flux of electrons is incident. So, in Eq. (1), A_p will be a function of V. If the sheath is thin compared to the probe radius, $r_p \gg \lambda_D$, the current is determined by the space charge–diode relationship, as discussed by Chen,[3] and the electron saturation region can be used to give a value for $n(kT_e)^{1/2}$. When the sheath is thick, $r_p \ll \lambda_D$, we can no longer consider that the particles entering the sheath are directed perfectly inward toward the probe. Instead the angular momentum and possible orbital motion of the electrons must be considered as an important limitation to the current collected. In this case, changing V will have an important effect on the attractive potential, and orbital motion can have a strong effect on the "saturation" current. As Chen[3] points out, in this case the electron saturation current will vary as V for spheres and $V^{1/2}$ for cylindrical probes. In practical cases, then, it may be quite difficult to use the electron saturation region to determine electron density. Figure 4 shows an experimental curve trace from the paper by Schoenberg[7] and illustrates the "nonflat" nature of the electron saturation region. Another point that must also be considered is that driving the probe into electron saturation may draw significant current from the plasma and cause an unacceptable perturbation of the properties of the plasma in the vicinity of the probe which will result in a reading that is not actually representative of the plasma properties in the absence of the probe.

Figure 4 does suggest, however, that region I, the ion saturation region, is reasonably well behaved. One would assume that for a probe with a large negative bias, all electrons would be repelled and the random ion current incident upon the sheath edge, as given in Eq. (2), would give the

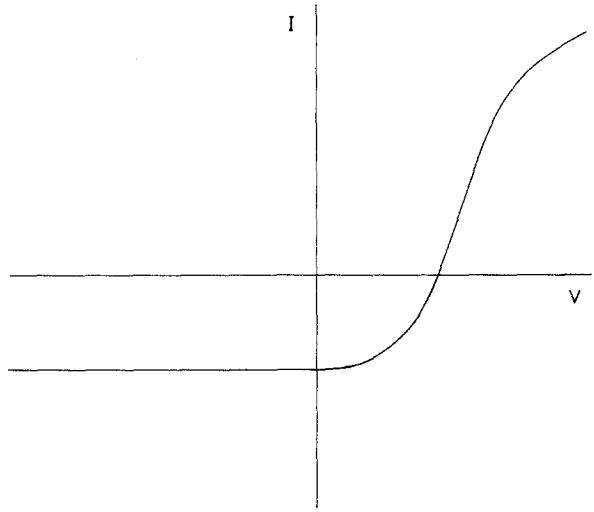


Fig. 4. Experimental cylindrical probe I–V characteristic. The axes are arbitrarily placed (adapted from Schoenberg[7]).

saturation current. Unfortunately, the assumption of a well-defined sheath edge poses a problem here. A careful solution of the bipolar diode corresponding to such a condition shows that the assumption of a discrete sheath edge is only tenable if the colder particles, ions in this case, enter the sheath with a finite initial velocity, v_0, determined by the Bohm criterion

$$(Mv_0^2/2kT_e)^{1/2} > \frac{1}{\sqrt{2}} \tag{8}$$

In other words, for the condition $T_e \gg T_+$, the ions must enter the sheath with an energy kT_e. The ions are accelerated through a "pre-sheath" to the required energy, and therefore the collected ion saturation current becomes, approximately,

$$I_+ = 0.605 A_p ne(kT_e/M)^{1/2} \tag{9}$$

This expression can then be used in conjunction with a measurement of T_e to deduce the plasma density, but the ion temperature essentially cannot be measured.

Because the idea of a well-defined sheath edge breaks down for positive ion collection, the "exact" treatment of ion collection at low pressures requires a careful consideration of ion orbits and a self-consistent solution of Poisson's equation. The artifice of a sharp boundary can, in fact, be completely removed by using a boundary-layer analysis. By necessity, then,

a rigorous approach to the positive ion collection characteristic results in very complex solutions with little intuitive insight. Equation (9) gives an order-of-magnitude check on plasma density, but more accuracy requires a careful matchup to theoretically generated curves such as those given by Bernstein and Rabinowitz[8] or Lam.[9] Refer to Appendix 4 of the monograph by Swift and Schwar[4] for an informative table of references on the theories of positive ion collection in low-pressure plasmas.

2.2. Measurement of Electron Energy Distribution Functions

In low-pressure molecular gas discharges characteristic of those encountered in plasma chemistry applications, it would be unusual to find an electron energy distribution that was Maxwellian. In this situation electron temperature no longer has a well-defined meaning and it is often desirable to deduce the distribution function both as an indicator of important processes occurring in plasma and for use in estimating critical reaction rates. The studies of the distribution function in CO_2 laser discharges at pressures of up to 10.0 torr by Bletzinger and Garscadden[10] indicate the importance of knowing the distribution function in such non-Maxwellian situations.

The previous section discussed how the isotropic distribution function can be derived, but it still ignores a very important fact. In electrically produced plasmas, especially at low pressures, the electron distribution function is anisotropic in geometrical space as well as in velocity space. This aspect of electrostatic probes has been studied in detail by Shkarofsky.[11] Figure 5 shows plane probe characteristics obtained in low-pressure (0.14 torr) nitrogen discharges for different orientations of the probe with respect to the electric field in the discharge. Clearly, the substantial differences in the curves can be related to the spatial anisotropy induced by the imposed electric field. It is possible to use the probe curves for different orientations to deduce the various components of the distribution function f_0, f_1, and f_2 that represent the first three terms of the usual spherical harmonic expansion of the distribution function:

$$f = f_0(v) + \bar{f}_1(v) \cdot \frac{\bar{v}}{v} + \bar{f}_2(v) : \bar{v}\bar{v}/v^2 + \cdots \tag{10}$$

Figure 6 shows the isotropic part of the distribution function, f_0, and the component directed along the electric field, f_1, that correspond to the probe curves shown in Fig. 5. The quantity f_0 is clearly non-Maxwellian and f_1 is of substantial magnitude.

As the pressure increases to the point where the mean free path approaches the size of the probe ($l \simeq r_p$), it becomes more difficult to deduce

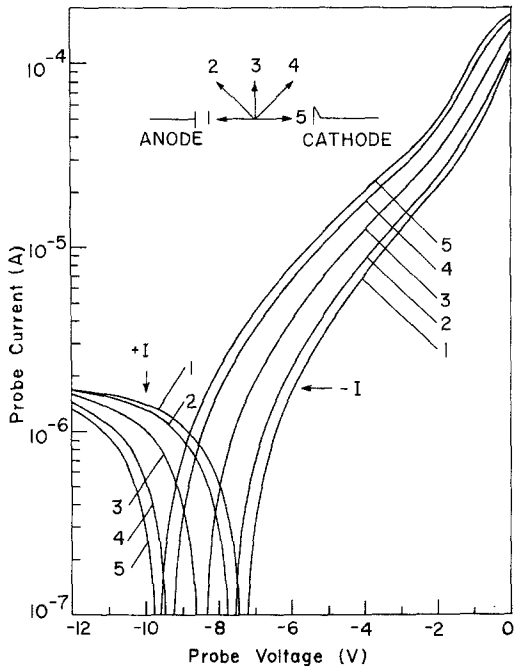


Fig. 5. Current–voltage characteristics of a plane probe for five orientations when $E = 4.5$ V/cm, fractional ionization $= 10^{-6}$, and $p = 0.14$ torr in nitrogen (adapted from Shkarofsky[11]).

the distribution function from the probe curve, since the distribution function at the sheath edge differs from that in the bulk plasma. It is still possible, however, to use an iterative procedure to deduce f_0 from the probe curve if $T_e \gg T_+$. At much higher pressures, in the continuum range, it is possible to deduce the probe characteristic if f_0 is known, but it is not possible to deduce the distribution function from the probe curve.

2.3. Double Probes

The probe shown in Fig. 2 makes measurements with reference to a large electrode in the plasma, which is often taken to be the anode of the main discharge or a metallic wall. In case there is no convenient reference electrode, such as in RF discharges, it is sometimes convenient to use a balanced floating double probe, as shown in Fig. 7. This measurement technique is also useful for transient plasma studies where the plasma potential varies with time. Since both probes are insulated from ground,

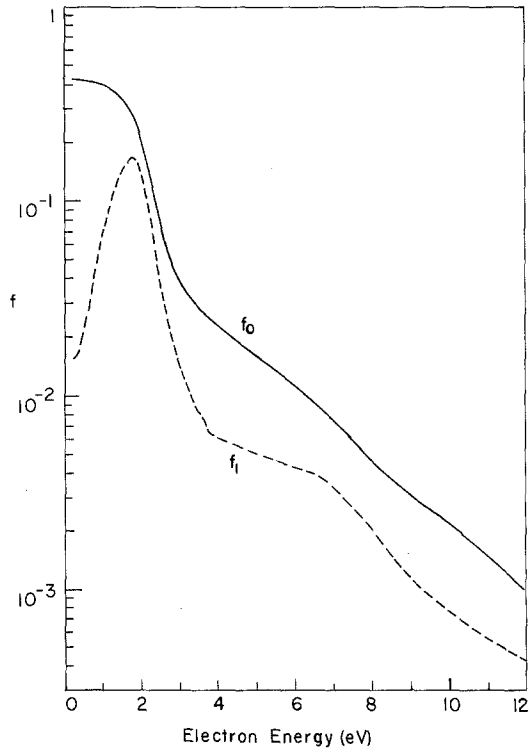


Fig. 6. Isotropic distribution function, f_0, and first-order term, f_1, corresponding to probe curves shown in Fig. 5 (adapted from Shkarofsky[11]).

the probes float with the plasma and are not affected by changes in the plasma potential.

The potential reached by each probe with respect to the plasma potential, V_p, is shown in Fig. 8. Both probes will be at a negative potential with respect to the plasma since no net current can flow to the system. In order to balance the circuit, the electron current flowing to probe 1 must be balanced by the ion current flowing to probe 2. There is an analog to two *p–n* junction devices connected back to back. The currents are therefore limited by the ion saturation currents for each probe. This ensures a minimal perturbation of the plasma by the probes. Under the assumption that the probes are identical and that there is no difference in the plasma potential between the two probe locations, the *V–I* characteristic is as shown in Fig. 9. When V is changed, both V_1 and V_2 change so that the electron current collected by one probe balances the ion current collected by the other.

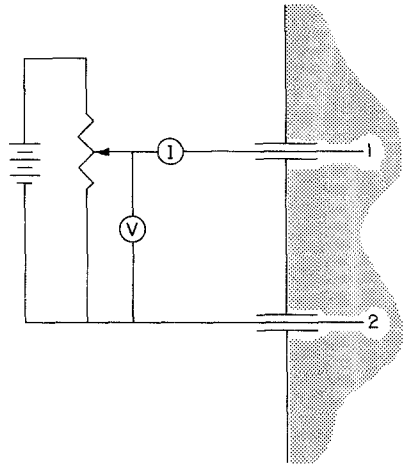


Fig. 7. Measuring circuit for a floating double probe.

When V is large positive, probe 2 is driven into ion saturation, I_{2+}, and V_1 will adjust itself to collect enough electron current to cancel the ion saturation current. In this case, the probe areas are the same, so the I–V curve is symmetric. If the areas are different, the saturation currents will scale accordingly. The V–I characteristic can be found by overlapping the ion saturation characteristic of each probe, as shown in Fig. 9. In the idealized case where the electron saturation regions are flat, the resulting expression for the current in terms of the voltage is

$$I = I_+ \tanh (eV/2kT_e) \tag{11}$$

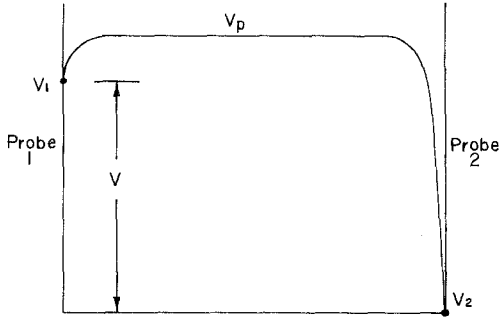


Fig. 8. Potential of the double probes with respect to the plasma potential.

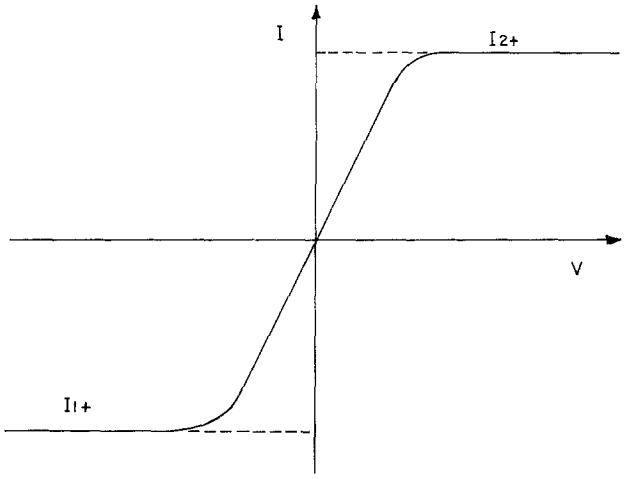


Fig. 9. Typical double-probe I–V characteristic.

The simplest approach to determining T_e from such a curve is to take the slope of the curve at the origin

$$\left.\frac{dI}{dV}\right|_{I=0} = \frac{e}{kT_e}\frac{I_+ I_{2+}}{I_{1+}+I_{2+}} \tag{12}$$

This approach is useful if the characteristics exhibit good saturation. Swift and Schwar[4] discuss the interpretation when the preceding condition does not hold. Once kT_e is known, the ion saturation current can then be used to compute the ion density.

One important property of the double probe, as can been seen from Fig. 8, is that only a small fraction of the energy distribution of the electrons in the plasma, the high-energy tail, is sampled by this probing method. Swift and Schwar[4] estimate that less than 14% of the random electron current is sampled by such a symmetrical system. Unless the electron energy distribution is truly Maxwellian, a somewhat rare situation, no knowledge of the electron temperature of the bulk of the electrons in the plasma can be deduced by the symmetric double probe.

2.4. Triple Probes

A variation of the floating probe method is the triple probe, as shown in Fig. 10. In this case, a third probe with a large surface area is inserted into the plasma. As before, the net current to the system is zero. The voltage–current relationship for probes 1 and 3 are measured, but the area

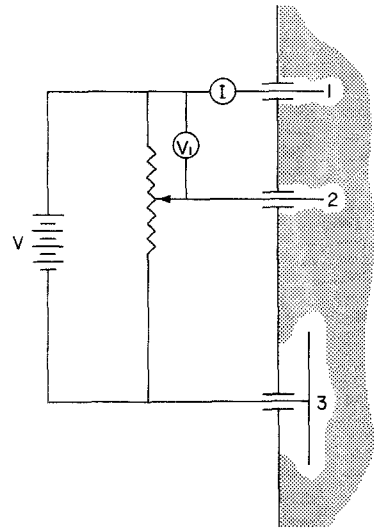


Fig. 10. Measuring circuit for a triple probe.

of probe 3 is made sufficiently large so that sufficient ion saturation current can flow to probe 3 to allow probe 1 to go into electron saturation. The probe characteristic is then essentially what would be expected for a single probe. The purpose of probe 2 is to establish the floating potential by setting the voltage divider so that no current flows to probe 2. In this way, the current to probe 1 can be determined directly in terms of the potential difference between probe 1 and the floating potential.

The obvious disadvantage of the triple probe is the necessity to insert more "probe area" into the plasma with the corresponding greater potential for significant perturbation of the plasma properties by the measuring scheme.

2.5. Emissive Probes

As was seen in the previous discussions, the plasma potential, V_p, is quite difficult to estimate from the probe characteristics, and yet it is a quantity of fundamental importance by itself as well as for the correct interpretation of probe characteristics. The plasma potential, V_p, is often estimated by taking the intersection of the electron saturation characteristic with the electron retarding characteristic. Yet, as can be expected from Fig. 3, this can give rise to a considerable uncertainty. Another approach to explicitly determine the plasma potential is to use an electron emissive

probe, typically made of a fine hairpin of tungsten wire heated to incandescence. When the probe is biased strongly positive with respect to the plasma, the electrons emitted will be returned to the probe and no difference should be observed between the characteristics of the probe when hot or cold. When the potential of the probe is negative with respect to the plasma potential, V_p, we will have a gas diode and the emitted electrons will flow across the sheath. A significant difference in the characteristic will then be seen for probe potentials $V < V_p$. As shown by Chen,[3] the intersection between the hot and cold probe characteristics will be a good indication of the plasma potential. Of more interest is the fact that the floating potential will approach the plasma potential as the probe emission increases. This was shown very nicely by Kemp and Sellen,[12] as in Fig. 11. It is not always possible, however, to approach the plasma potential since the emission is always limited by space-charge effects.

3. MEASUREMENT CIRCUITS

The circuits shown in Figs. 2, 7, and 10 show how probe measurements could be basically made. Actually these circuits are far too simplistic, as might be expected. One of the fundamental reasons why point-by-point measurements are not made is that probes are subject to contamination and drift of characteristics, as will be discussed later, so measurements are normally made on a short time scale, or on a "dynamic" basis. The whole question of the dynamic characteristics of a probe are therefore very important and will be addressed shortly.

In virtually all measurement circuits, the voltage applied to the probe is swept over the entire measurement range in a relatively short period of

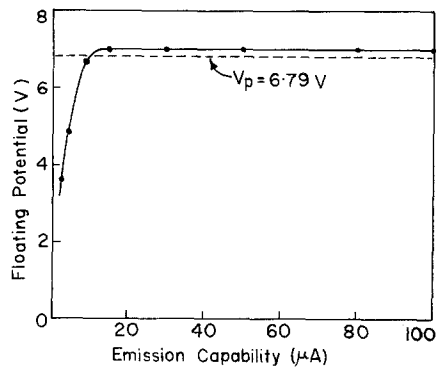


Fig. 11. Floating potential as a function of emission capability for an electron emissive probe (adapted from Kemp and Sellen[12]).

time, or the probe voltage is pulsed to a specified value and the current measured. With modern programmable power supplies and signal digitizers, it is quite conceivable to use a measuring circuit essentially as simple as those shown in Figs. 2, 7, and 10. The data can then be processed by computer with good curve fitting procedures and the essential parameters automatically deduced. In most cases, however, the measurement circuits are designed to directly process the data with analog circuits for direct display or subsequent computerized data collection and manipulation. For instance, one common approach is to linearly sweep the probe voltage and to use several differentiators and logarithmic amplifiers to directly display quantities that can be related to electron energy distribution, number density, and electron temperature. The real-time analog computer developed by Nordlund and Breaux[13] is a good example. A block diagram of their system is shown in Fig. 12. The doubly differentiated signal, e_2, is related to the electron energy distribution function, f_e, which can be obtained directly by multiplying e_2 by $\sqrt{V-V_p}$ and by using the proper scaling. Multiplying by $\sqrt{V-V_p}$ converts the distribution function given in Eq. (5) to a more commonly used normalization. The value of $\sqrt{V-V_p}$ is found by assuming that $V = V_p$ when the second derivative, e_2, is zero. By integrating e_3, which is proportional to f, a quantity proportional to the electron density is found. Once again, the second derivative, e_2, can lead to a measure of electron temperature, T_e, by differentiation of the log of e_2. This circuit was designed to sweep 10 times per second, and signal averaging can be used to improve accuracy and eliminate noise and fluctuations provided that the plasma is relatively well behaved.

Schoenberg[7] has used a measurement system similar to that of Nordlund and Breaux, except that he digitally records the probe characteristic

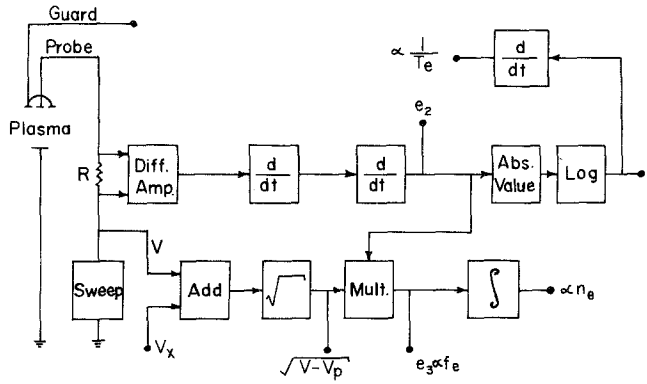


Fig. 12. Block diagram of an analog plasma diagnostic system (adapted from Nordlund and Breaux[13]).

as well as the first and second derivatives and uses a computer to process and interpret those signals. This system has been designed to acquire data in a range of 0.1 to 1 ms so that it can be used on transient or noisy plasmas. The ability to ensemble average over several data sets improves the resolution of the measurements.

As in virtually all probe measuring systems, prior to the sweep, Schoenberg,[7] as well as Nordlund and Breaux,[13] applied a large voltage to the probe so that it will be cleaned by bombardment.

The differentiating circuits described so far have used the traditional operational amplifier configurations. A slightly different approach has been used by Shimizu and Amemiya.[14] They particularly wanted to make very rapid measurements in noisy plasmas, so they used a linear sweep and two delay elements in conjunction with operational amplifiers to determine the second derivative of the probe characteristic. In this case, it is actually a second difference, rather than a second derivative.

Floating-probe measuring systems provide an additional problem. Since both probes float, and hence are useful in time-varying plasmas, or plasmas with rapidly changing plasma potentials, it is necessary to "isolate" the circuit from ground as well as possible. Mossack and Rusbridge,[15] as well as Hoegger and Bulliard,[16] have taken advantage of using optically isolated elements for this purpose. Hoegger and Bulliard[16] drove their circuit with a high-performance DC–DC converter with 40 pF stray capacitance, high isolation, and less than 50 pF capacitance to ground. Mossack and Rusbridge's[15] application of the optical isolators was in interfacing signals to an analog-to-digital converter in a computer-controlled data-acquisition scheme.

3.1. A.C. Measuring Techniques

One of the most widely used methods for "preprocessing" probe characteristics is to superimpose a small a.c. voltage on top of the d.c. probe bias voltage and look at the a.c. component of the probe current at the second harmonic of the drive frequency. Superimposing an a.c. voltage, v_p, on the d.c. probe bias, V_p, causes an increase in the d.c. probe current as well as the generation of fundamental frequency plus harmonic constituents in the a.c. probe current. In the notation used by Swift and Schwar,[4] if the probe current–voltage relationship can be expressed by

$$I = F(V_p) \tag{13}$$

then the incremental change in the d.c. probe current is

$$\Delta I_{d.c.} = (v_p^2/4)(d^2F/dV_p^2) + (v_p^4/64)(d^4F/dV_p^4) + \cdots \tag{14}$$

and the second harmonic component of the a.c. current is

$$\Delta I_{\text{a.c.},2} = (v_p^2/4)(d^2F/dV_p^2) + (v_p^4/48)(d^4F/dV_p^2) + \cdots \quad (15)$$

If v_p is small, the first terms in both expansions will dominate. In terms of measurement case, $\Delta I_{\text{d.c.}}$ will represent a small change, in general, of the total d.c. current and will be difficult to measure if there is any noise or fluctuations in the plasma. On the other hand, the second-harmonic component can be carefully filtered and phase-sensitive detection can be used to eliminate extraneous signals and allow measurements to be made even in relatively noisy environments. The second-harmonic measurement then gives a signal which can be directly related to the electron energy distribution function.

Branner, Friar, and Medicus[17] as well as Richards, Lloyd, and Jones,[18] have described this second-harmonic measurement technique in their papers. One obvious experimental constraint that they point out is the need for a very pure sinusoidal driving signal and linear amplifying circuits to ensure that second-harmonic generation only occurs in the plasma, and not in the measuring circuitry.

Amemiya and Shimizu[19] have also pointed out an important constraint on the frequency of the a.c. signal used for the second-harmonic measurement system. As was discussed previously, it is often desirable to make a rapid sweep of the d.c. probe bias voltage over its entire range. If meaningful a.c. measurements are to be made, the period of the a.c. signal must be much smaller than the period of the sweep. This ensures that a number of a.c. cycles can be measured in each meaningful interval of the d.c. voltage. Obviously, then, it is desirable to go to higher a.c. signal frequencies. The limitation that Amemiya and Shimizu[19] found is that the a.c. signal frequency should always be less than one-tenth of the plasma frequency. Above this value, we approach resonance probe effects, and errors will occur in the measurements.

3.2. Plasma Resonance Probes

Equation (14) showed that the d.c. probe current varied as an a.c. signal was applied. Figure 13 shows how this incremental current varies as the frequency of the signal increases. Over a substantial frequency range, $\Delta I_{\text{d.c.}}$ is independent of frequency. Above a certain frequency, as discussed earlier, the currents are very sensitive functions of frequency. At low frequencies, the electrons and ions can respond quickly enough to always be in equilibrium with the applied field. In the region where the incremental current peaks, the period of oscillation becomes comparable to the ion transit time through the sheath. Above that frequency, the electrons

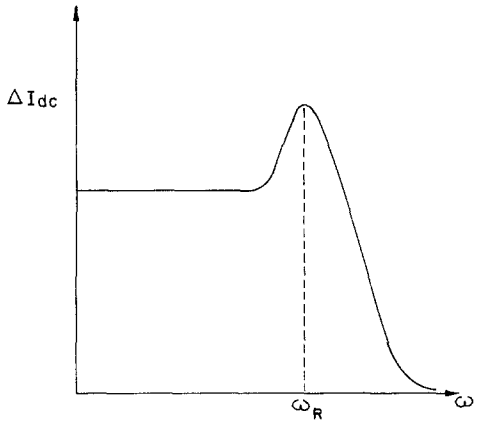


Fig. 13. Incremental d.c. current as a function of the frequency of a superimposed a.c. signal.

experience many a.c. periods as they traverse the sheath, and the effect of the a.c. field averages out. It was originally assumed that the resonant peak occurred when $\omega_R = \omega_p$, the plasma frequency. This was an erroneous interpretation, and it was later shown that the resonant peak was due to a large field generated when the plasma and the sheath formed a series resonant circuit as shown in Fig. 14. For a spherical probe in an unbounded plasma, the resonant frequency, ω_p, can be approximated to within 10% by

$$\omega_R^2 = \omega_p^2/(1 + R/k\lambda_D) \tag{16}$$

where λ_D is the electronic Debye length, R is the probe radius, and $k \simeq 5$.

3.3. Probe Dynamic Effects

The resonance effect previously described represents one dynamic effect that occurs when an a.c. field is applied whose period coincides with the transit time across the sheath. There are additional effects which become particularly evident when short pulses are applied to probes. Bills, Holt, and McClure,[20] as well as Oskam, Carlson, and Okuda,[21] have provided some valuable studies of these effects. There are generally two concerns to be addressed at this time. The first is the capacitive effect, and the resulting displacement current, due to the sheath capacitance and to the parallel capacitance of the glass insulator that usually surrounds a probe. The second is the time that it takes the electrons and ions to reach a new equilibrium position as the probe potential is changed.

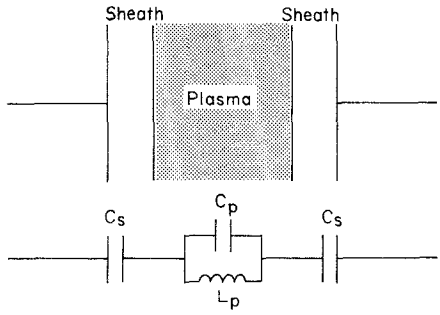


Fig. 14. Simple model of a plasma between metallic electrodes and the a.c. equivalent circuit.

The behavior of the plasma sheath capacitance is in itself interesting, but the stray capacitance introduced by the glass insulation on the probe introduces a displacement current that can cause errors in interpretation. Chen[3] showed a circuit in which a dummy probe was used with a differential input current detector in an attempt to cancel out the stray capacitance. This is still subject to error, however, since the dummy probe is not surrounded by a plasma. Carlson, Okuda, and Oskam[22] have resolved this problem by redesigning the probe to incorporate a metallic shield, as shown in Fig. 15. The grounded shield eliminates the spurious capacitive effects, since the displacement currents due to these effects no longer will run through a grounded measuring resistor. The purely capacitive properties of the probe can be related to the sheath conditions as discussed by Crawford and Grard.[23]

The time constant involved in the redistribution of electron and ion charge as the probe potential is varied is determined largely by the time it takes the ions to move, since the electrons are much more mobile. At low pressures, where the ions move in a collision-free environment, the frequency of response would be expected to be of the order of the ion plasma frequency. At higher pressures, the ion motion will be essentially collision dominated and the time constant will vary inversely with the plasma density and the ion mobility, as shown by Okuda, Carlson, and Oskam[24] In this case, it may take many microseconds before an equilibrium distribution, and probe current, is reached.

In the paper by Bills, Holt, and McClure,[20] an interesting facet of this problem was shown. When the probe is rapidly pulsed *above* the plasma potential, and only then, the probe current exhibits an overshoot on the leading edge. This shows some promise as a method for determining the plasma potential.

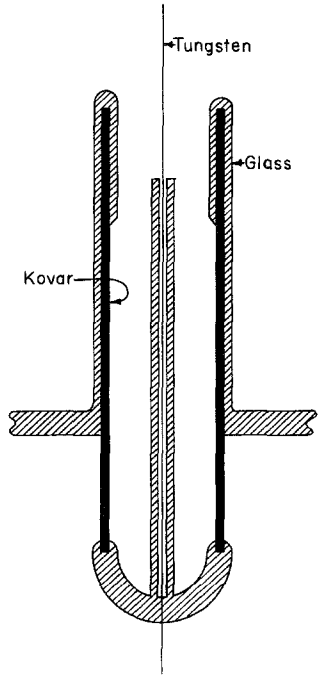


Fig. 15. A shielded Langmuir probe (adapted from Carlson, Okuda, and Oskam[22]).

3.4. Probe measurements in Noisy or Striating Plasmas

The subject of probe measurements in oscillating plasmas has been reviewed by Chen.[3] The results, as shown by Garscadden and Emeleus,[25] indicate that electron temperature measurements are usually still valid if the probe is kept in the electron retarding region, but the probe curves are shifted and the transition to electron saturation is broadened so that the plasma potential as inferred from the break in the probe curve will be in error and the electron concentration will be underestimated.

Discharges with striations are cases of special interest in plasma chemistry situations, since electronegative gases often give rise to striated discharges.[26] In such cases, it is necessary to use a phase reference based on the striation in order to obtain detailed temporal information on the behavior of the plasma,[27,28] otherwise the probe curves will be broadened.

4. PROBES IN HIGH-PRESSURE PLASMAS

In the foregoing discussion, the emphasis has been on low-pressure plasmas that are appropriate for sputtering, etching, deposition, and a

number of laser media. There are also quite a number of important plasma chemistry applications where the electron temperatures are less than 20,000 K and the gas pressure is essentially atmospheric (760 torr). These applications encompass MHD devices, flames, arcs, plasma jets, and shock-heated plasmas. In such species, it is usual for the mean free path to be small with respect to the sheath thickness and possibly less than the thickness of the probe ($l \ll r_p, \lambda_D$). In this probe regime, collisional effects dominate and continuum approaches are necessary.

One of the basic problems with probes at high pressures is that since $l \ll r_p$, the plasma density is depleted near the probe and the electrons and ions must diffuse from the bulk of the plasma to the sheath edge. In this case the probe influence extends some distance into the plasma where ion production is important. In order to reduce the carrier drain and the perturbation of the plasma, these probes are normally operated in the strongly negative regime.

With a negatively biased probe, it is usual to assume that the electrons have Boltzmann distribution:

$$n_e = n \exp(eV/kT_e) \tag{17}$$

and the ion motion is determined by diffusion in the bulk plasma and is mobility controlled in the sheath so that

$$J_+ = -eD_+ \nabla n_+ - n_+ e\mu_+ \nabla V \tag{18}$$

where D_+ is the ion diffusion coefficient, μ_+ is the ion mobility, and V is the potential.

The theory for cylindrical probes is problematic at high pressures, so most of the theory is given for spherical probes only. The works by Wasserstrom, Su, and Probstein,[29] Cohen,[30] and Su and Lam[31] are most often referred to. Appendix 5 of Swift and Schwar[4] gives a table of high-pressure probe theories and their ranges of solution.

4.1. Stationary and Flowing High-Pressure Plasmas

If a spherical probe is biased strongly negatively, it should be possible to relate the asymptotic saturation current to the bulk plasma density. Su and Lam[31] have derived such an expression in the continuum limit for a stationary probe as given in Eq. (19) (MKS units)

$$I_+ = (4\pi\varepsilon_0)^{1/4}(Vr_p)^{1/2}\mu_+(4\pi nkT_e)^{3/4}(1 + T_+/T_e)^{3/4} \tag{19}$$

This expression still requires a knowledge of T_e that can be found by operating the probe in the electron retarding region as was discussed previously. In particular, the double-probe method is still useful.[4] The

quantity T_+ still cannot be deduced with any accuracy, but the ion temperature and the neutral gas temperature should be approximately the same.

Equation (19) is valid for stationary high-pressure plasmas, and this implies a particularly stringent condition that the plasma flow velocity be less than 0.1–1 m/sec. This is not at all appropriate in a number of important plasma chemistry applications involving MHD, flames, arcs, or flowing plasma reactors. In these situations, the probe equations must be rederived for a flowing plasma, and an interesting simplification occurs. The determining factor for the flow of particles to the probe is now the flow velocity, V_f, rather than the temperatures of the electrons and ions, and T_e and T_+ drop out of the equation.[6]

Clements and Smy[32–34] have derived a member of very useful expressions for the asymptotic current to a negatively biased probe under flowing conditions. Specifically, this is the condition that the plasma flow velocity exceeds the ambipolar ion diffusion velocity, or the electric Reynolds number, $R_e > 1$, where

$$R_e = \frac{V_f 2 r_p}{\mu_+ (kT_e/e)} \tag{20}$$

For a spherical probe with a thin sheath for a flowing continuum plasma, the asymptotic ion current[32] is

$$I_+ = 12.13(\mu_+ \varepsilon_0)^{1/4} (neV_f)^{3/4} V^{1/2} r_p^{5/4} \tag{21}$$

and this equation gives good agreement with experiments.

For a spherical probe with a thick sheath,[6] we have

$$I_+ \simeq \pi (neV_f)^{0.6} (6\varepsilon_0 \mu_+)^{0.4} (Vr_p)^{0.8} \tag{22}$$

and this equation gives agreement within 50% with experimental values.

For a cylindrical probe with a thick sheath,[34] we have

$$I_+ = \frac{2(\pi \mu_+ \varepsilon_0)^{1/3} (neV_f V)^{2/3}}{[\log (I_+/2neV_f r_p)]^{2/3}} \tag{23}$$

again, with agreement within ±50%.

5. PROBLEMS IN PROBE MEASUREMENTS

A number of difficulties in the use of probes and the interpretation of probe data have already been pointed out. It is now appropriate to mention additional limitations and sources of error.

1. The probe must not significantly perturb the plasma to be measured.[35,36] I use the word significantly here, since any metallic object

inserted into a plasma will obviously cause some perturbation. Two things are essential. The distance of penetration of the probe field into the plasma must be small with respect to distances over which significant changes in plasma parameters, such as plasma potential, occur. Also, the loss of carriers to the probe must be much less than the loss due to other effects. Obviously, the probe must be physically small. Planar probes are generally too large, and very fine cylindrical and spherical probes are used. Even then, it is extremely unlikely that probes can be used in plasmas with large field gradients such as in cathode regions or in coronas, even if elementary precautions are used to ensure that the fields are not shorted out. In low-pressure plasmas, the requirement that the carrier drain to the probe be small is usually attainable unless there are strong magnetic fields. In such cases, the probe can influence the plasma over significant distances. The same is also true for higher-pressure plasmas, since the probe influence is felt into the ionization region.

Naturally, a probe will perturb the neutral gas temperature in the vicinity. If there is a significant coupling between neutral or ion temperatures and the plasma properties, as occurs at high pressures and in chemically reacting plasmas, such as flames, then the effect can be significant and the use of probes under such conditions must be approached with caution.

It must also be kept in mind that a probe to which an a.c. signal, a pulse, or a swept d.c. voltage is applied acts as an antenna. As a result, it is quite possible that the signal applied to the probe can excite a number of instabilities and fluctuations in a plasma, such as striations. Any such induced plasma behavior is usually strongly coupled to the plasma parameters and can cause a significant perturbation as well as very distorted and unrepresentative measurements.

2. The chemical nature of the probe material, its surface, and supports plays a very important role. In the first place, it is not unusual to have the probe be made of a different metal (usually tungsten or molybdenum) than the reference electrode (anode) in the plasma. There is then a contact potential that must be taken into effect as an addition to or substraction from the voltage applied to the probe.

An additional source of error, especially at low frequencies in rare gas plasmas, is photoemission from the probe. Just as photoemission can play a significant role in cathode phenomena, so too can it contribute to the "ion current" of the probe, and in largely an unknown way. It has even been estimated that photoelectric emission may constitute up to 10% of the ion saturation current. Metastable atoms, especially in rare gases, can also cause electron emission from the probe. In molecular gases, neither metastables nor photoelectric effects would be expected to be significant.

3. In electronegative gases, the contribution due to negative ions must be taken into account. Boyd and Thompson[37] have extended probe theory to consider negative ions. Thompson[38] has also shown the influence of negative ions on diffusion in electronegative plasmas. In many molecular gases, especially oxygen or halogen-bearing gases, a large concentration of negative ions can be expected. This is usually not a significant problem unless the negative-ion concentration is quite large with respect to the electrons. This is due to the fact that the low negative-ion temperature and large mass make the negative ion contribution to the current small for reasonable negative-ion concentrations. The general criterion for neglecting the negative ions is

$$\frac{N_-}{n_e} \ll \left(\frac{M_- T_e}{m T_-}\right)^{1/2} \tag{24}$$

where N_-, M_-, and T_- are the negative ion density, mass, and temperature, respectively, and n_e is the electron density.

4. The conditions must be such that the electron energy distribution reaching the probe is representative of that in the bulk of the plasma. This is somewhat related to the perturbation of the plasmas by the probe, but is not an identical condition. The loss of particles to the probe may be relatively small, but when the pressure is high, collisions will alter the energy distribution of the particles approaching the probe so that the probe measurements do not reflect bulk plasma electron energy distributions. At low pressure, Waymouth[39] showed that the fast electrons collected by the probe will deplete the plasma and cause a distortion of the distribution in the plasma. This is, indeed, a significant perturbation of the plasma by the probe.

5. In RF plasmas, the probe circuit must be decoupled from the RF field that sustains the plasma. Double probes are often isolated by putting RF chokes in the probe leads and by putting a capacitor between probes to eliminate a.c. fields between the probes. If a common high-amplitude RF field is coupled to the probes, the nonlinear sheath characteristic can give a rectified high bias on the probe[40] that will drive the probes into saturation, as shown by Stampa and Wulf.[41]

5.1. Probe Contamination

In chemically reactive plasmas, probe contamination seems to be a primary limiting factor on probe usefulness. If the probe is coated or contaminated by plasma constituents, the work function change during measurements will cause a significant shift in the V–I curve. Even contamination of part of the probe surface can cause a distortion of the characteris-

tic. This effect has been shown by a number of authors.[42–46] One specific effect that is seen when contamination is a problem is that the probe behavior becomes current dependent, particularly if there is a buildup of dielectric impurities, and the *V–I* characteristic exhibits a hysteresis curve as shown by D'Arcy.[45] Wunderer[47] has designed a measurement circuit that uses a referenee probe in the plasma to try to eliminate the effect of work function fluctuations.

Since probe contamination is a very important problem in plasma chemistry applications, and the potential to lay down polymer coatings on the probe is obvious in many applications, it is necessary to properly clean the probe prior to measurement. This is usually done by applying a large negative voltage to the probes so that ion bombardment will occur. The probe characteristic is then swept in a time short compared to contamination times. Dynamic measurements are a virtual necessity.

It is, of course, important to have some method of knowing when the probe is clean and of monitoring the buildup of deposited layers. The studies by Olson and Nordlund[48] and Olson[49] have shown that the susceptance–voltage characteristic of a probe is a sensitive indicator of the presence of insulating material on the probe surface. In fact, the susceptance–voltage curve for an insulator-contaminated probe has a peak in the vicinity of the plasma potential which disappears upon removal of the insulating material by sputtering. Of some concern is Olson's observation that it can take quite a long period of time to actually clean a probe. At least his method allows the "clean" condition to be determined and monitored. In addition, if the primary purpose of the discharge is deposition, Olson's method would permit an on-line evaluation of the deposition rate.

As a possible bright note, Oliver, Clements, and Smy[50] and Clements[51] have observed that RF probes can be quite insensitive to surface contamination and oxide layers for measurements of electron density and electron temperature. It may be possible to generalize and extend this approach.

6. COMPARISON OF PROBE DIAGNOSTICS WITH OTHER PLASMA MEASUREMENT TECHNIQUES

Unfortunately, there are all too few comparisons of plasma parameters measured by probes and by other independent techniques. Clements and Smy[32–34] have used microwave measurements to validate their probe data at high pressures. At low pressures, the measurements of Kinderdyk and Van Eck[52] showed that probe measurements gave values 30% lower than that deduced from microwave cavity and microwave interferometer studies in low-pressure helium discharges. Similar measurements by Schulz and

Brown[53] have shown a generally good agreement between probe and microwave measurements in a pressure range of 0.05 to 6 torr, which covers the transition from collisionless probes to collision-dominated probes. Self and Shih[54] have compared probes and microwaves with reasonably good agreement at low pressures. Sicha, Gajdusek, and Veprek[55] have found similar agreement for studies in neon at 1.8 torr.

7. CONCLUSIONS

In contrast to the implied ease of use that a cursory understanding of probes might suggest, the proper experimental technique and theoretical basis for interpreting probe measurements remains a very challenging problem. At low pressures, probe theory is relatively well understood but very complex and often difficult to implement. At high pressures, the theory is not well formulated, but electron densities can be measured with reasonable accuracy. Realistic measurement circuits are of considerable complexity. Numerous complicating effects and sources of error can arise due to the natural conditions occurring in complex plasmas as well as due to the perturbations caused by the intrusion of a probe.

This litany of negatives does not mean that probes cannot provide useful information. Indeed, data on plasma potentials and electron energy distribution functions, when possible, must use some type of probe. It is important to realize the inherent problems and potential inaccuracies of probes, however, and to consider data obtained by such means to be approximate and rarely, if ever, conclusive. Probes can yield useful data, particularly if it is used on a comparative basis. Users of probes, however, must be fully cognizant of the limitations of the theory and the sources of error in the experiment.

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